Weak solution of the merged mathematical equations of the polluted atmosphere

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Abstract

Considered as a geophysical fluid, the polluted atmosphere shares the shallow domain characteristics with other natural large-scale fluids such as seas and oceans. This means that its domain is excessively greater horizontally than in the vertical dimension, leading to the classic hydrostatic approximation of the Navier-Stokes equations. The authors of the [6] article have proved a convergence theorem for this model with respect to the ocean, without considering pollution effects. In this present work we provide a generalization of their result translated to the atmosphere, extending the fluid velocity equations with an additional convection-diffusion equation representing pollutants in the atmosphere.

Keywords: Navier-Stokes equations, shallow domains, pollution evolution equation, hydrostatic approximation, compactness, weak solutions.

Contents

1 Introduction

The key aspects of capturing the dynamics of either water flow in oceanography or atmospheric changes in meteorology are the following two fundamental concepts that underlie many modern asymptotic models aiming to describe them. The first one is that both above mentioned phenomena can be viewed as a fluid dynamics process, and as such, they are best described by the Navier-Stokes equations. The second is the notion of the so-called shallow domains. This latter is a widely used concept in the field of large-scale geophysical fluids and its main idea is to take advantage of their approximately flat structure when modeling them with the Navier-Stokes equations. Essentially, the basic principle is to exploit the fact that when we are working on domains such as the ocean or the atmosphere, we are working on an "almost" two-dimensional set, and thus we can significantly simplify the model in the vertical direction, or even eliminate the third dimension and arrive to a limit model in two dimensions. A common practice for example is to depth-integrate the Navier-Stokes equations for big lakes and seas and obtain the shallow water equations. Our approach is different, we keep the vertical velocity and will arrive to a three dimensional limit model. We use the

$$\epsilon = \frac{\text{characteristic depth}}{\text{characteristic width}}$$

aspect ratio and the fact that, as the above described general ideas suggest, it is natural to consider asymptotic models and to take the limit model as ϵ goes to zero.

In this paper we consider the phenomena of a contaminant being emitted into, mixing with, and being deposited from the homogeneous and clean air. This topic is found in the intersection of heavily applied fields such as mathematical fluid dynamics, atmosphere modeling, and meteorology; for a profound insight see e.g. the books [5], [15] and [18].

Our focus is to create a self-contained model describing the atmospheric flow combined with the pollution effects and to investigate the asymptotic behavior of the weak solutions of this model. The model itself, new to our knowledge, utilizes classical concepts as its foundation and it merges them with relatively new or more specific ideas. On the one hand, we use general and well-known approaches such as the wind flow being modeled by the incompressible anisotropic Navier-Stokes equations, and using a convection-diffusion equation to describe the pollution effect. On the other hand, the model also incorporates notions like taking advantage of a downwind-direction matching coordinate system ([9]), using a Gaussian pulse as a source term ([22]), and rescaling the diffusivity parameters according to the largely different horizontal and vertical scales (similarly to the way the viscosity terms are scaled in [6]).

Specifically, as for the limit behavior of the weak solutions of the model we construct, we show that the weak solutions of this model converge to a weak solution of the hydrostatic limit model — in other words, taking the limit we arrive to a justification of the hydrostatic model for the case of the polluted atmosphere, where, instead of the originally used wind momentum equation, the hydrostatic air pressure assumption is used to describe the process in the vertical dimension. Since the polluted atmosphere as a merged physical phenomena can naturally be seen as the air-analogous version of the salty sea, with this convergence result we are providing an expansion of the main theorem of [6] in the sense that we obtain a result describing the atmospheric flow combined with pollution effects, while the authors of [6] provide a similar result for the ocean excluding salinity.

This paper is organized in the following way. In section 2 we describe the physical model of the polluted atmosphere and the scaling leading to the hydrostatic equations. In section 3 we introduce the function spaces we work in, and we present the weak formulation and the main theorem. The main theorem's proof is described in section 4, and a generalized model is introduced in chapter 5. A few concluding remarks can be found in section 6, finally in the appendix of the article we present an existence result for the hydrostatic equations of the polluted atmosphere.

We end up this section with some notations we will use through the paper.

1.1 Notations

a) $\epsilon =$ the aspect ratio

b) $\nu =$ viscosity

c) $\Omega_\epsilon, \Omega =$ the original and the rescaled (ϵ-independent) domain, respectively

d) $\nabla = (\partial_x, \partial_y, \partial_z)$ the gradient vector

e) $\nabla_\nu = (\nu_x^{1/2}\partial_x, \nu_y^{1/2}\partial_y, \nu_z^{1/2}\partial_z)$

f) $\Delta_\nu = \nu_x\partial^2_{xx} + \nu_y\partial^2_{yy} + \nu_z\partial^2_{zz}$ the anisotropic Laplacian

g) $\mathbf{g}$ = represents the force due to gravity, which also includes the centrifugal effect

h) φ = the gravity potential term, i.e. $\mathbf{g} = \nabla\varphi$

i) $\mathbf{w}$ represents the earth rotation angular speed, $l(y)$ = latitude

j) $\alpha = 2f\sin(l(x_2))$, $\beta = 2f\cos(l(x_2))$

k) $(\cdot,\cdot)$ = the scalar product in $L^2(\Omega)^d$ or the duality $L^p(\Omega)$, $L^{p'}(\Omega)$

l) $<\cdot,\cdot>_\Gamma$= the scalar product or duality on the boundary curve Γ $(H^{-1/2}(\Gamma), H^{1/2}(\Gamma))$

m) $\mathbf{u}_H$ = the horizontal velocity components (u_1, u_2)

n) $b(\mathbf{u}_H) = \alpha(-u_2, u_1)$

2 The polluted atmosphere model

We begin by choosing a coordinate system, set the domain we start with and an appropriate set of equations describing both atmosphere motions and the effect of pollution. We ignore large scale effects such as the curvature of the Earth. For describing the effect of pollution we use a continuity equation written in Cartesian coordinates for the concentration.

We fix the local Cartesian coordinate system to be adjusted to the downwind direction, i.e. z is the vertical direction, while x is the downwind, and y is the crosswind direction. It is a legitimate assumption if we are considering a time interval which is not too long in the sense that it is reasonable to assume that we have a permanent, relatively stable main wind direction.

The domain we consider is a local slice of the atmosphere filled with homogeneous air that we assume to be incompressible, and a pollutant of concentration C, therefore it is defined by the set

$$\Omega_\epsilon = \{(x, y, z) \in \mathbb{R}^3; (x, y) \in \omega, 0 < z < \epsilon h(x, y)\}, \tag{1}$$

where ω is a Lipschitz-domain in $\mathbb{R}^2$, h is a nonnegative Lipschitzian application, and ϵ incorporates the shallowness of the domain.

The equations governing the air velocity $\mathbf{v}$ and pressure q in the atmosphere are the incompressible Navier-Stokes equations, in which we use different viscosities according to vertical or horizontal directions, and the density is taken to be identically equal to one:

$$\partial_t \mathbf{v} + (\mathbf{v}\cdot\nabla)\mathbf{v} - \Delta_\nu \mathbf{v} + \nabla q + 2\mathbf{w}\times\mathbf{v} = \mathbf{g} \text{ in } \Omega_\epsilon \times (0,T) \tag{2}$$

$$\nabla\cdot\mathbf{v} = 0 \text{ in } \Omega_\epsilon \times (0,T) \tag{3}$$

Contaminants emitted into the atmosphere are transported by the wind, mixed and dispersed by turbulent behavior, and deposited onto the ground by gravity. The dispersion of pollutants is commonly modeled by a diffusive equation of the form

$$\partial_t C + \mathbf{v}\cdot\nabla C = K_x \frac{\partial^2 C}{\partial x^2} + K_y \frac{\partial^2 C}{\partial y^2} + K_z \frac{\partial^2 C}{\partial z^2} + S, \tag{4}$$

where C is the pollution concentration.

In this equation S represents the sources and sinks of the pollutant in the atmosphere i.e. its emission, removal from the atmosphere by dry and wet deposition, and chemical reactions. We will use the approach suggested by [22] and use what is called a parametrized nascent delta function (a Gaussian impulse, specifically) to describe the source term S.

Let us consider a gas plume with intensity I originating at emission coordinates $\mathbf{x}_s$ beginning at time t_s. At the beginning we assume zero concentration. An abrupt gas emission is modeled by a source term represented as a product of an approximated delta function $\delta_\kappa(\mathbf{x}-\mathbf{x}_s)$ (centered at $\mathbf{x}_s$ with a constant parameter κ) and a Heaviside function s (switching from 0 to I at time t_s.) In fact, this source term models the continuous emission of the gas at constant rate and fixed location. Basically $s(t;t_s)$ is a function which activates the source, namely $s(t;t_s) := IH(t-t_s)$ where $H(\cdot)$ is the Heaviside step function and $0 \le t_s \le T$. We arrive to the complete source term model of the form

$$S_\kappa(t, t_s, x, x_s) = s(t;t_s)\delta_\kappa(\mathbf{x}-\mathbf{x_s}), \tag{5}$$

while in the limit model the source term S naturally stands for the δ distribution.

Now, we perform a vertical scaling to make the domain independent of ϵ.

$$x = x_1, \quad y = x_2, \quad z = \epsilon x_3.$$

So we have the following new, ϵ - independent domain

$$\Omega = \{(x_1,x_2,x_3)\in\mathbb{R}^3; (x_1,x_2)\in\omega, 0 < x_3 < h(x_1,x_2)\}.$$

The corresponding scaling regarding the kinematic and viscosity terms are

$$v_1 = u_1^\epsilon, \quad v_2 = u_2^\epsilon, \quad v_3 = \epsilon u_3^\epsilon, \quad p = p^\epsilon,$$
$$\nu_x = \nu_1, \quad \nu_y = \nu_2, \quad \nu_z = \epsilon^2 \nu_3,$$

where the later embodies the different eddy viscosity hypothesis that we mentioned earlier, and p is the pressure term including the gravity potential φ, i.e., $p = q - \varphi$. Basically it can be heuristically deduced from the viscosity dimensions: typical length^2/typical time, more details on the subject can be found in [7], [6].

Finally, we use an analogous rescaling methodology for the diffusivity constants in (4),

$$K_y = K_2, \quad K_z = \epsilon^2 K_3.$$

As a second step of the rescaling process, we also exploit the fact that we chose the coordinate system in a way so that the x axis matches the downwind direction. In this case we use an additional scaling equation, which is different in nature from the previous ones in the sense that it is not derived from the shallowness of the domain but rather suggested by the fact that in the downwind direction it is natural to assume that the diffusion is negligible compared to downwind transport, i.e.

$$\left| u_1 \partial_{x_1} C \right| \gg \left| K_x \frac{\partial^2 C}{\partial x_1^2} \right| \tag{6}$$

which leads us to using

$$K_x = \epsilon K_1.$$

Finally, as we mentioned before, we use a Gaussian approximated delta function to represent the source, so it is convenient to reformulate (5) in terms of ϵ, which is the standard deviation parameter of the distribution describing the source — since later we will take the limit as ϵ goes to zero, it will result in having the δ distribution representing the hydrostatic source term, which is a physically natural choice. In other words, we choose κ to be ϵ and the following function will give shape to the source term:

$$\delta_\epsilon(\mathbf{x} - \mathbf{x_s}) = \gamma \epsilon^{-1} e^{\frac{-(\mathbf{x}-\mathbf{x_s})^2}{\epsilon^2}};$$

and finally

$$S_\epsilon = IH(t-t_s)\gamma\epsilon^{-1}e^{\frac{-(\mathbf{x}-\mathbf{x_s})^2}{\epsilon^2}}, \tag{7}$$

where γ is a normalizing constant.

Putting this condition and the previously described rescaling terms together to the original equations, we arrive to the merged anisotropic equations of the polluted atmosphere above inland surface:

$$\partial_t u_1^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_1^\epsilon - \Delta_\nu u_1^\epsilon - \alpha u_2^\epsilon + \epsilon\beta u_3^\epsilon + \partial_1 p^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{8}$$

$$\partial_t u_2^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_2^\epsilon - \Delta_\nu u_2^\epsilon + \alpha u_1^\epsilon + \partial_2 p^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{9}$$

$$\epsilon^2\{\partial_t u_3^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon - \Delta_\nu u_3^\epsilon\} - \epsilon\beta u_1^\epsilon + \partial_3 p^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{10}$$

$$\nabla \cdot \mathbf{u}^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{11}$$

$$\partial_t C^\epsilon + \mathbf{u}^\epsilon \cdot \nabla C^\epsilon = \epsilon K_1 \frac{\partial^2 C^\epsilon}{\partial x_1^2} + K_2 \frac{\partial^2 C^\epsilon}{\partial x_2^2} + K_3 \frac{\partial^2 C^\epsilon}{\partial x_3^2} + S_\epsilon \text{ in } \Omega \times (0,T) \tag{12}$$

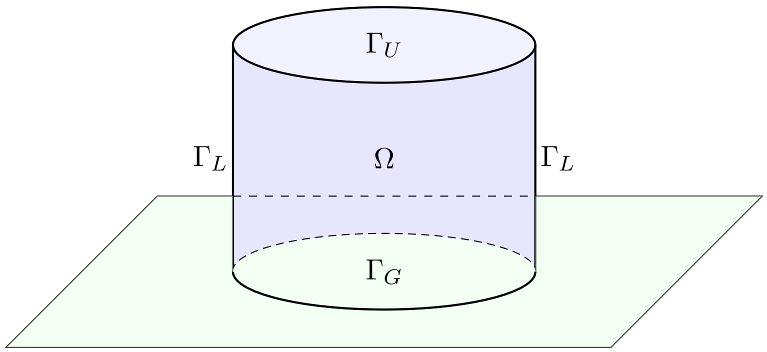
The initial condition for the velocity and the pollution concentration are

$$\mathbf{u}^\epsilon(\cdot, t=0) = \mathbf{u}_0, \quad C^\epsilon(\cdot, t=0) = C_0 \quad \text{ in } \Omega. \tag{13}$$

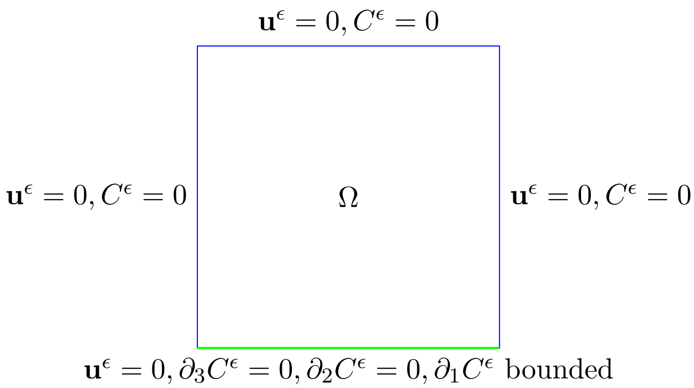
To make the above set of equations complete, we need to assign the boundary conditions.

Concerning the pollution concentration it is vital to choose boundary conditions that are physically as accurate as possible and at the same time not unnecessarily complicated. As we can see in many related papers, in the case of LAMs (limited area models) finding the right boundary conditions can be a nontrivial task.

In order to describe the boundary condition in a precise way we need to divide $\Gamma = \partial\Omega$ as follows. The upper boundary of the Ω domain is denoted by Γ_U, the lateral boundary is Γ_L, and the lower boundary of the domain is Γ_G. As we mentioned before, the case we are investigating is that of inland domains Ω, i.e. the case of landscapes and cities with no significant nearby water surface. This is very important since the boundary



conditions for both wind velocity and pollution concentration in the air are different depending whether we are above solid ground or water surface. The boundary conditions we choose to use in the inland scenario are the following:



- The upper boundary Γ_U of the domain we are considering is chosen to be at the isobar representing the planetary boundary layer ([5]), therefore we assume

$$\mathbf{u}_H^\epsilon = 0, \quad u_3^\epsilon = 0, \quad C^\epsilon = 0 \quad \text{on } \Gamma_U \times (0,T) \tag{14}$$

- On Γ_L we use the same boundary conditions as on the upper level, i.e. we have

$$\mathbf{u}_H^\epsilon = 0, \quad u_3^\epsilon = 0, \quad C^\epsilon = 0 \quad \text{on } \Gamma_L \times (0,T). \tag{15}$$

Remark 1. *If we wanted to use $C^\epsilon(x,y,z) = c$ with some c constant value, we could make a change of variables distracting that background flow and arrive back to the same mathematical problem with C being zero.*

- In this model we neglect topological variations, and the $z = 0$ plane represents the lower boundary. On the ground level we have

$$\mathbf{u}_H^\epsilon = 0, \quad u_3^\epsilon = 0 \quad \text{on } \Gamma_G \times (0,T). \tag{16}$$

  This is the mathematical way of expressing no-slip boundary conditions for the horizontal wind speed and the impervious nature of the ground with respect to the wind. As for the concentration, the way we can choose ground level boundary conditions is the following. Firstly, we assume that the pollution is not absorbed by the ground, so we use

$$\partial_3 C^\epsilon = 0 \text{ on } \Gamma_G$$

  (see also for example [21]). For the horizontal partial derivatives of the pollution, we can separate downwind and crosswind direction boundary conditions. In the crosswind direction we use

$$\partial_2 C^\epsilon = 0 \text{ on } \Gamma_G,$$

  while in the downwind (stronger) direction our assumption is more relaxed, we only assume that $\partial_1 C^\epsilon$ is finite in the boundary norm, i.e. in $L^2(0,T;\Gamma_G)$.

Finally, the boundary conditions to describe the anisotropic system can be summarized as follows:

$$\begin{aligned}
&\mathbf{u}^\epsilon = 0 \text{ on } \Gamma \times (0,T), \\
&C^\epsilon = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0,T), \\
&\partial_2 C^\epsilon = \partial_3 C^\epsilon = 0 \text{ on } \Gamma_G \times (0,T), \quad \partial_1 C^\epsilon \in L^2 L^2(0,T;\Gamma_G)
\end{aligned} \tag{17}$$

Note that we have a domain where the lower and upper boundaries represent a physically existent layer (i.e. the ground and the planetary boundary layer), but outside the lateral boundary the physical world continues without influence and without any particular external condition. On boundaries of such nature we have to apply OBCs (open boundary conditions), which is not a trivial task, see for example [20]. We need to avoid any spurious reflection or constraint which is unnatural, but at the same time we want the model to remain mathematically manageable.

If we assume $\mathbf{u}^\epsilon = O(1)$, then neglecting the ϵ^2 and ϵ in the anisotropic equations, we formally arrive to the hydrostatic Navier-Stokes equations (i.e., primitive equations) combined with pollution.

$$\partial_t u_1 + \mathbf{u} \cdot \nabla u_1 - \Delta_\nu u_1 - \alpha u_2 + \partial_1 p = 0 \text{ in } \Omega \times (0, T) \tag{18}$$

$$\partial_t u_2 + \mathbf{u} \cdot \nabla u_2 - \Delta_\nu u_2 - \alpha u_1 + \partial_2 p = 0 \text{ in } \Omega \times (0, T) \tag{19}$$

$$\partial_3 p = 0 \text{ in } \Omega \times (0, T) \tag{20}$$

$$\nabla \cdot \mathbf{u} = 0 \text{ in } \Omega \times (0, T) \tag{21}$$

$$\partial_t C + \mathbf{u} \cdot \nabla C = K_2 \frac{\partial^2 C}{\partial x_2^2} + K_3 \frac{\partial^2 C}{\partial x_3^2} + S \text{ in } \Omega \times (0, T) \tag{22}$$

$$u_1 = u_2 = u_3 n_3 = 0 \text{ on } \Gamma \times (0, T) \tag{23}$$

$$u_i(\cdot, t = 0) = u_{0i}, C(\cdot, t = 0) = C_0 \text{ in } \Omega, i = 1{,}2 \tag{24}$$

$$\begin{aligned} &C = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0, T), \\ &\partial_2 C = \partial_3 C = 0 \text{ on } \Gamma_G \times (0, T), \partial_1 C \in L^2 L^2(0, T; \Gamma_G) \end{aligned} \tag{25}$$

Remark 2. *Here S stands for the δ limit distribution. We keep the notation S for generality, since as we will see later, instead of the approximated delta functions and the delta distribution, it is possible to use any sort of "sufficiently smooth" source term (which can be independent from ϵ in the anisotropic case as well) that is bounded in $L^2_t L^2_x$.*

3 Weak formulation and Main Result

In this section we describe the weak formulation of the merged equations in both the anisotropic and hydrostatic case we are going to use in the paper, and we state our main result.

We need to define the following spaces:

$$C_\Gamma^\infty(\Omega) = \{\varphi \in C^\infty(\bar{\Omega}); \varphi = 0 \text{ on some neighbourhood of } \Gamma\}$$

$$H_\Gamma^k(\Omega) = \overline{C_\Gamma^\infty(\Omega)}^{H^k(\Omega)} = \{v \in H^k(\Omega); v = 0 \text{ on } \Gamma\}$$

$$\mathbf{V} = \{\mathbf{v} \in H_\Gamma^1(\Omega) \times H_\Gamma^1(\Omega) \times H_0^1(\Omega); \nabla \cdot \mathbf{v} = 0 \text{ in } \Omega\}$$

$$H(\partial_3, \Omega) = \{v \in L^2(\Omega); \partial_3 v \in L^2(\Omega)\}$$

$$H_0(\partial_3, \Omega) = \overline{C_0^\infty(\Omega)}^{H(\partial_3,\Omega)} = \{v \in H(\partial_3, \Omega); v n_3 = 0 \text{ on } \Gamma\}$$

$$\mathbf{W} = \{\mathbf{u} \in H_\Gamma^1(\Omega) \times H_\Gamma^1(\Omega) \times H_0(\partial_3, \Omega); \nabla \cdot \mathbf{u} = 0 \text{ in } \Omega\}$$

Then the weak formulation of the hydrostatic system (18) - (25) takes the following form.

Definition 1. *Find $(\mathbf{u}, C)$ such that $\mathbf{u} = (\mathbf{u}_H, u_3) \in L^2(0, T; \mathbf{W})$ with $\mathbf{u}_H \in L^\infty(0, T; L^2(\Omega)^2)$, and $C \in L_t^\infty L_x^2$ with $\partial_2 C, \partial_3 C \in L_t^2 L_x^2$, such that*

$$\begin{aligned}
&\int_0^T \Big[-(\mathbf{u}_H, \partial_t \tilde{\mathbf{u}}_H) + (\nabla_\nu \mathbf{u}_H, \nabla_\nu \tilde{\mathbf{u}}_H) - (\mathbf{u}_H, (\mathbf{u} \cdot \nabla)\tilde{\mathbf{u}}_H) + (b(\mathbf{u}_H), \tilde{\mathbf{u}}_H) \Big] \,\mathrm{d}t \\
&+ \int_0^T \Big[-(C, \partial_t \tilde{C}) - (\mathbf{u}C, \nabla \tilde{C}) + K_2(\partial_2 C, \partial_2 \tilde{C}) + K_3(\partial_3 C, \partial_3 \tilde{C}) \Big] \,\mathrm{d}t \\
&= -(\mathbf{u}_{0H}, \tilde{\mathbf{u}}_H(0)) - (C(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C}) \,\mathrm{d}t
\end{aligned} \tag{26}$$

for all $(\tilde{\mathbf{u}}, \tilde{C})$ with $\tilde{\mathbf{u}} = (\tilde{\mathbf{u}}_H, \tilde{u}_3) \in H^1(0, T, \mathbf{W})$, with $\tilde{\mathbf{u}}_H(T) = 0$ and $\partial_3 \tilde{\mathbf{u}}_H \in L^\infty(0, T; L^3(\Omega)^2)$ and $\tilde{C}$ with $\tilde{C} \in L^2(0, T, H_\Gamma^3(\Omega))$, $\tilde{C} \in H^1(0, T, L^2)$, $\tilde{C}(T) = 0$.

Note that the term $\int_0^T (S, \tilde{C})\, \mathrm{d}t$ makes sense since in this case S denotes the delta distribution and we have

$$(S, \tilde{C}) = \int_\Omega \delta(\mathbf{x} - \mathbf{x}_s)\tilde{C}(\mathbf{x})\, \mathrm{d}\mathbf{x} = \tilde{C}(\mathbf{x}_s).$$

Remark 3. *If S is chosen to be different from the limit of the Gaussian nascent delta function, then it is easy to see that the sufficient assumption we have to make on a general S hydrostatic pollution function is that it should be in $L_t^2 L_x^2$.*

The existence of weak solution for the hydrostatic system (18) - (25) can be proved in the spirit of [19]. In appendix A we will outline the main steps of the proof and the differences compared to the ocean model.

The weak form of the anisotropic system is as follows.

Definition 2. *Find $\mathbf{u}^\epsilon = (\mathbf{u}_H, u_3^\epsilon) \in L^2(0,T;\mathbf{V}) \cap L^\infty(0,T;L^2(\Omega)^3)$ and $C^\epsilon \in L^\infty(0,T;L^2(\Omega)) \cap L^2(0,T;H^1(\Omega))$, such that*

$$\begin{aligned}
&\int_0^T \Big[-(\mathbf{u}_H^\epsilon, \partial_t \tilde{\mathbf{u}}_H) + (\nabla_\nu \mathbf{u}_H^\epsilon, \nabla_\nu \tilde{\mathbf{u}}_H) - (\mathbf{u}_H^\epsilon, (\mathbf{u}^\epsilon \cdot \nabla)\tilde{\mathbf{u}}_H) + (b(\mathbf{u}_H^\epsilon), \tilde{\mathbf{u}}_H)\Big]\, \mathrm{d}t \\
&+ \epsilon\beta \int_0^T \Big[(u_3^\epsilon, \tilde{u}_1) - (u_1^\epsilon, \tilde{u}_3)\Big]\, \mathrm{d}t \\
&+ \epsilon^2 \int_0^T \Big[-(u_3^\epsilon, \partial_t \tilde{u}_3) + (\mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon, \tilde{u}_3) + (\nabla_\nu u_3^\epsilon, \nabla_\nu \tilde{u}_3)\Big]\, \mathrm{d}t \\
&+ \int_0^T \Big[-(C^\epsilon, \partial_t \tilde{C}) - (\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}) + \\
&\epsilon K_1(\partial_1 C^\epsilon, \partial_1 \tilde{C}) - \epsilon K_1 \big[(\partial_1 C^\epsilon)\tilde{C}\big]_{\Gamma_G} + K_2(\partial_2 C^\epsilon, \partial_2 \tilde{C}) + K_3(\partial_3 C^\epsilon, \partial_3 \tilde{C})\Big]\, \mathrm{d}t = \\
&= -(\mathbf{u}_{0H}^\epsilon, \tilde{\mathbf{u}}_H(0)) - \epsilon^2 (u_{03}^\epsilon, \tilde{u}_3(0)) - (C^\epsilon(0), \tilde{C}(0)) + \int_0^T (S^\epsilon, \tilde{C})\, \mathrm{d}t
\end{aligned} \tag{27}$$

for all $\tilde{\mathbf{u}} = (\tilde{\mathbf{u}}_H, \tilde{u}_3) \in H^1(0,T;\mathbf{V})$ with $\tilde{\mathbf{u}}(T) = 0$ and $\tilde{C}$ such that $\tilde{C} \in H^1(0,T,H_\Gamma^3(\Omega))$ and $\tilde{C}(T) = 0$.

Note that since in our case the source term S_ϵ is given by (5), the last term in (27) is well defined. If we use a source function of a different and more general form, it needs to satisfy the $L_t^2 L_x^2$ boundedness criteria for the weak formulation to make sense.

We will omit the proof of the existence of weak solutions (see Definition 2.) for the anisotropic system (8) - (17) since it follows by standard finite-dimensional Galerkin approximation.

Now we are ready to state our main result.

Theorem 1. *Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0, \mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; let $C_0 \in L^2(\Omega)$, $C_0 = 0$ on $\partial\Omega$, and assume that the boundary conditions (17) hold; then as the aspect ratio ϵ tends to zero, any weak solution $(\mathbf{u}^\epsilon, C^\epsilon)$ of the anisotropic equations (8) - (12) converge to a weak solution $(\mathbf{u}, C)$ of the hydrostatic equations of the polluted atmosphere (18) - (22).*

4 Proof of the main theorem

This section is devoted to the proof of Theorem 1. In the extended case with the pollution equation the proof relies on a priori estimates gained from the energy inequality and, as we will see this is sufficient to pass to the limit in the linear terms. Concerning the nonlinear terms that do not include pollution we will follow [6]. What we need to work on is how to handle the nonlinear term including the new function C^ϵ. The challenge caused by this nonlinear term will be met by proving that a compactness result from [12] is applicable to handle it.

4.1 Energy inequality

As we mentioned before, one of the most fundamental tools in this proof is the energy inequality. If we calculate it for the anisotropic system, we obtain that for a.e. $t \in [0, T]$ the following holds:

$$\begin{aligned}
&\frac{1}{2}(\left\|\mathbf{u}_H^\epsilon(t)\right\|_{L^2}^2 + \epsilon^2\left\|u_3^\epsilon(t)\right\|_{L^2}^2 + \left\|C^\epsilon(t)\right\|_{L^2}^2) + \int_0^t \{\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2\|\nabla_\nu u_3^\epsilon\|_{L^2}^2\}\,\mathrm{d}\tau \\
&+ \int_0^t \{\epsilon K_1\|\partial_1 C^\epsilon\|_{L^2}^2 + K_2\|\partial_2 C^\epsilon\|_{L^2}^2 + K_3\|\partial_3 C^\epsilon\|_{L^2}^2\} d\tau \\
&\leq \int_0^t (S^\epsilon, C^\epsilon)\,\mathrm{d}\tau + \frac{1}{2}(\left\|\mathbf{u}_H^\epsilon(0)\right\|_{L^2}^2 + \epsilon^2\left\|u_3^\epsilon(0)\right\|_{L^2}^2 + \|C_0^\epsilon\|_{L^2}^2) \\
&+ \int_0^t \epsilon K_1 \langle \partial_1 C^\epsilon, C^\epsilon\rangle_{\Gamma_G}\,\mathrm{d}\tau \\
&= I_1 + I_2 + I_3.
\end{aligned} \tag{28}$$

From the hypothesis on the initial data and since the S^ϵ source term is in $L^\infty_x(\Omega)$, I_1 and I_2 are bounded. The only term we need to work on further is I_3.

By using the Holder inequality and the generalized Young inequality for any $\zeta > 0$, we can estimate this term in the following way.

$$
\begin{aligned}
I_3 &= \int_0^t \epsilon K_1 \langle \partial_1 C^\epsilon, C^\epsilon \rangle_{\Gamma_G} \,\mathrm{d}\tau \le K_1 \int_0^t \sqrt{\epsilon} \|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)} \sqrt{\epsilon} \|C^\epsilon\|_{L^2(\Gamma_G)} \\
&\le \frac{1}{\zeta} K_1 \frac{\epsilon}{2} \int_0^t \|\partial_1 C^\epsilon\|^2_{L^2(\Gamma_G)} + \zeta K_1 \int_0^t \frac{\epsilon}{2} \|C^\epsilon\|^2_{L^2(\Gamma_G)} \\
&\le \frac{1}{\zeta} K_1 \frac{\epsilon}{2} \int_0^t \|\partial_1 C^\epsilon\|^2_{L^2(\Gamma_G)} + \zeta \frac{\epsilon}{2} K_1 c_T \int_0^t \|C^\epsilon\|^2_{W^{1,2}(\Omega)} \\
&\le \frac{1}{\zeta} K_1 \frac{\epsilon}{2} \int_0^t \|\partial_1 C^\epsilon\|^2_{L^2(\Gamma_G)} + \zeta \frac{\epsilon}{2} K_1 c_T (c_S + 1) \int_0^t \|\nabla C^\epsilon\|^2_{L^2(\Omega)} \,\mathrm{d}\tau,
\end{aligned}
\tag{29}
$$

where c_T and c_S are the constants provided by the trace theorem and Sobolev embeddings, and the boundary term is bounded because of the a priori assumptions made on the boundary conditions.

The energy inequality (28) now takes the form

$$
\begin{aligned}
&\frac{1}{2} (\|\mathbf{u}^\epsilon_H(t)\|^2_{L^2} + \epsilon^2 \|u^\epsilon_3(t)\|^2_{L^2} + \|C^\epsilon(t)\|^2_{L^2}) + \int_0^t \{\|\nabla_\nu \mathbf{u}^\epsilon_H\|^2_{L^2} + \epsilon^2 \|\nabla_\nu u^\epsilon_3\|^2_{L^2}\} \,\mathrm{d}\tau \\
&+ \int_0^t \{(\epsilon K_1 - \zeta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) \|\partial_1 C^\epsilon\|^2_{L^2} + (K_2 - \zeta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) \|\partial_2 C^\epsilon\|^2_{L^2} \\
&+ (K_3 - \zeta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) \|\partial_3 C^\epsilon\|^2_{L^2}\} \,\mathrm{d}\tau \\
&\le \frac{1}{\zeta} K_1 \frac{\epsilon}{2} \int_0^t \|\partial_1 C^\epsilon\|^2_{L^2(\Gamma_G)} + \frac{1}{2} (\|\mathbf{u}^\epsilon_H(0)\|^2_{L^2} + \epsilon^2 \|u^\epsilon_3(0)\|^2_{L^2} + \|C^\epsilon(0)\|^2_{L^2}) \\
&+ \int_0^t (S^\epsilon, C^\epsilon) \,\mathrm{d}\tau.
\end{aligned}
\tag{30}
$$

The right hand side of the energy inequality is now apparently bounded, and it is easy to see that the coefficients of the pollution terms are positive, since the ζ constant of the generalized Young inequality can be freely chosen in a way such that the requirements $\epsilon K_1 (1 - \zeta \frac{1}{2} c_T (c_S + 1)) > 0, (K_2 - \zeta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) > 0, (K_3 - \zeta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) > 0$ are all satisfied.

4.2 A priori estimates and weak convergence

From the final form of the energy inequality we obtain uniform a priori estimates that we summarize in the following proposition.

Proposition 1. *Let $u_1^\epsilon, u_2^\epsilon, u_3^\epsilon$ and C^ϵ be the weak solutions of the system (8) - (12) in the sense of Definition 2, assuming that the boundary conditions (17) hold. Then the sequences*

$$\begin{aligned}
&u_1^\epsilon, u_2^\epsilon, \epsilon u_3^\epsilon, C^\epsilon && \textit{are bounded in } L^\infty(0,T;L^2(\Omega)), && (31)\\
&u_3^\epsilon && \textit{is bounded in } L^2(0,T;L^2(\Omega)), && (32)\\
&u_1^\epsilon, u_2^\epsilon, \epsilon u_3^\epsilon && \textit{are bounded in } L^2(0,T;H^1(\Omega)), && (33)\\
&\sqrt{\epsilon}\partial_1 C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon && \textit{are bounded in } L^2(0,T;L^2(\Omega)). && (34)
\end{aligned}$$

Proof. It is straightforward that (31), (33) and (34) follow directly by the energy inequality (30). On the other hand, (32) can be verified by using the divergence free condition and that as a consequence, we have $\partial_3 u_3 = -\partial_1 u_1 - \partial_2 u_2$ bounded in $L_t^2 L_x^2$. The bound holds for u_3 itself as well, owing to the Poincare inequality in the vertical direction, and using that u_3 vanishes on the boundary. $\square$

As a consequence, up to subsequences still denoted by the same way, we have the following weak convergence results.

$$\begin{aligned}
&\mathbf{u}_H^\epsilon \rightharpoonup \mathbf{u}_H && \text{weakly in } L_t^\infty L_x^2 \cap L_t^2 H_x^1, && (35)\\
&\epsilon u_3^\epsilon \rightharpoonup f && \text{weakly for some limit element } f \text{ in } L_t^\infty L_x^2 \cap L_t^2 H_x^1, && (36)\\
&u_3^\epsilon \rightharpoonup u_3 && \text{weakly in } L_t^2 L_x^2, && (37)\\
&C^\epsilon \rightharpoonup C && \text{weakly in } L_t^\infty L_x^2, && (38)\\
&\partial_2 C^\epsilon \rightharpoonup \partial_2 C && \text{weakly in } L_t^2 L_x^2, && (39)\\
&\partial_3 C^\epsilon \rightharpoonup \partial_3 C && \text{weakly in } L_t^2 L_x^2, && (40)\\
&\sqrt{\epsilon}\partial_1 C^\epsilon \rightharpoonup e && \text{weakly for some limit element } e \text{ in } L_t^2 L_x^2. && (41)
\end{aligned}$$

4.3 Passing to the limit

In order to conclude the proof of Theorem 1 we pass to the limit in the weak formulation (27). To take the limit for the linear velocity terms that do not include the concentration function C^ϵ we can use (35), (36) and (37). As for the nonlinear terms, the convergence can be shown using a compactness theorem by [17]. The details are omitted since they are analogous to those described in [6].

For the linear terms including the concentration C^ϵ we can directly apply the weak convergence results of (38) - (41).

By using (38), (39) and (40) we get

$$\int_0^T (C^\epsilon, \frac{\partial \tilde{C}}{\partial t}) \mathrm{dt} \to \int_0^T (C, \frac{\partial \tilde{C}}{\partial t})\, \mathrm{d}t,$$

$$\int_0^T (\partial_2 C^\epsilon, f) + (\partial_3 C^\epsilon, f) \mathrm{dt} \to \int_0^T (\partial_2 C, f) + (\partial_3 C, f)\, \mathrm{d}t \qquad \forall f \in L^2_t L^2_x.$$

For the next term we have $\epsilon K_1 \int_0^T (\partial_1 C^\epsilon, \partial_1 \tilde{C}) = \sqrt{\epsilon} K_1 \int_0^T (\sqrt{\epsilon}\partial_1 C^\epsilon, \partial_1 \tilde{C})$, which goes to 0 as $\epsilon \to 0$ because of the Cauchy-Schwartz inequality, since we have $\sqrt{\epsilon}\partial_1 C^\epsilon$ bounded in $L^2_t L^2_x$, and $\tilde{C}$ is a test function.

The $\epsilon K_1 \int_0^T \left[(\partial_1 C^\epsilon)\tilde{C}\right]_{\Gamma_G} \mathrm{d}t$ line integral vanishes as $\epsilon \to 0$ since by the Cauchy-Schwartz equation we can estimate the integral by two bounded terms ($\partial_1 C^\epsilon$ was assumed to be finite in norm on the lower boundary).

For the source term, by applying standard results for distribution functions, we get $(S^\epsilon, \tilde{C}) \to (S, \tilde{C})$.

Finally we have to deal with the nonlinear term $(\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C})$.

By analyzing this nonlinear term in a more precise way we see that $(u_i C^\epsilon, \partial_i \tilde{C})$ for $i = 1, 2$ is weakly convergent since from [6] we have the strong convergence of the horizontal velocities, while in the case of $i = 3$ this property does not hold for u_3 and from the energy bound we do not have more than the weak convergence of C^ϵ .

In this case the classical Aubin-Lions lemma can not be applied since it would require for C^ϵ to be in a compactly embedded space such as $L^2 H^1$, which is not the case here because we only know boundedness results for $\sqrt{\epsilon}\partial_1 C$, but not for $\partial_1 C$.

Therefore we will use the following compactness result from [12].

Lemma 1. *Let g^ϵ, h^ϵ converge weakly to g, h respectively in $L^{p_1}(0, T; L^{p_2}(\Omega))$, $L^{q_1}(0, T; L^{q_2}(\Omega))$ where $1 \leq p_1, p_2 \leq +\infty$,*

$$\frac{1}{p_1} + \frac{1}{q_1} = \frac{1}{p_2} + \frac{1}{q_2} = 1.$$

We assume in addition that

$\dfrac{\partial g^\epsilon}{\partial t}$ *is bounded in $L^1(0, T; W^{-m,1}(\Omega))$ for some $m \geq 0$ independent of ϵ,* *and*

$$\left\| h^\epsilon - h^\epsilon(\cdot + \xi, t) \right\|_{L^{q_1}(0,T;L^{q_2}(\Omega))} \to 0 \ \textit{as}\ |\xi| \to 0, \ \textit{uniformly in}\ \epsilon.$$

Then, $g^\epsilon h^\epsilon$ converges to gh (in the sense of distributions on $\Omega \times (0,T)$).

What we will show in the following is that for $h^\epsilon = u_i^\epsilon$ and $g^\epsilon = C^\epsilon$ the assumptions of Lemma 1 are satisfied, which means that $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$ in the sense of distribution.

From (38) we have that $C^\epsilon \rightharpoonup C$ in $L_t^\infty L_x^2$, and since we are on a finite domain, the same convergence holds in $L_t^2 L_x^2$. Hence we can choose $q_1 = q_2 = p_1 = p_2 = 2$.

To estimate the time derivative of C^ϵ we use (12), that since u^ϵ is divergence-free can be rewritten as

$$\frac{\partial C^\epsilon}{\partial t} = -\nabla \cdot (\mathbf{u}^\epsilon C^\epsilon) + \sqrt{\epsilon} K_1 \partial_1 \partial_1 \sqrt{\epsilon} C^\epsilon + K_2 \partial_2 \partial_2 C^\epsilon + K_3 \partial_3 \partial_3 C^\epsilon - S^\epsilon$$

Now, the S^ϵ Gaussian source term is smooth; from (31), (32) we get $\nabla \cdot (\mathbf{u}^\epsilon C^\epsilon) \in W_x^{-1,1}$ and from (34) we have that the remaining terms are in $W_x^{-1,2}$.

Hence we can conclude that

$$\frac{\partial C}{\partial t} \in L_t^1 W_x^{-1,1}.$$

The property that

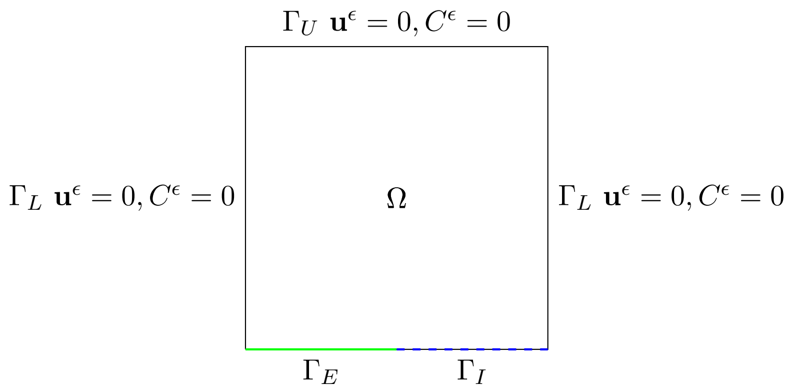
$$\left\| u_i^\epsilon - u_i^\epsilon(\cdot + \xi, t) \right\| \to 0 \text{ as } |\xi| \to 0$$

follows since $u_i^\epsilon \in L_t^2 L_x^2$.

Now we can apply Lemma 1 with $g^\epsilon = C^\epsilon, h^\epsilon = u^\epsilon, q_1 = q_2 = p_1 = p_2 = 2, m = -1$ and we finally arrive to

$$\int_0^T (\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C})\, \mathrm{d}t \to \int_0^T (\mathbf{u} C, \nabla \tilde{C})\, \mathrm{d}t,$$

which, combined with the previously described linear terms, concludes the proof of the main theorem.



5 Convection dominated coastal model

We now present a generalized model, which is an extension of the previous inland model by adapting the model for seaside conditions.

On the ground layer we naturally separate the boundary into two different parts in this model, $\Gamma_G = \Gamma_E \cup \Gamma_I$; Γ_E denoting the part of the boundary above the ground, and Γ_I denoting the section that is interacting with the sea. While above ground parts it is reasonable to use no-slip conditions, above the ocean it is natural to assume that the wind has an interaction with the waves, there is no such rigidness as in the ground case. For this cause we will include θ, which represents the wind traction on the ocean surface. Analogously, we need to make similar changes for the pollution concentration as well, above sea territory it is not natural to use $\partial_3 C = 0$, since the particles can now sink under the boundary level.

On Γ_E we have the same boundary conditions as in the inland case, $\mathbf{u}^\epsilon = 0$, and for the concentration we have $\partial_3 C^\epsilon = \partial_2 C^\epsilon = 0$, and $\partial_1 C^\epsilon$ being bounded. On Γ_I on the other hand we have $\nu_3 \partial_3 \mathbf{u}_H^\epsilon = \theta_H, u_3 = 0$ with assuming a finite wind energy, i.e. the function θ being in $L^2(0, T; H^{-1/2}(\Gamma_I))$. For the pollution concentration we use the same approach and assume that the functions $\partial_1 C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$ are equal to some sufficiently regular, ϵ-independent functions defined on the respective boundary. More accurately, we assume that we have a $C^\epsilon_{\Gamma_G} = \sigma$ sediment function on the lower boundary, where $\sigma \in L^2(0, T; H^1(\Gamma_G))$. If the partial derivatives of σ are denoted by σ_1, σ_2 and σ_3, we have $\partial_1 C^\epsilon = \sigma_1$ on Γ_G, and $\partial_2 C^\epsilon = \sigma_2$, $\partial_3 C^\epsilon = \sigma_3$ on Γ_I. (Of course σ has to be a function such that we have σ_2 and σ_3 equal to zero on the Γ_E part).

In what follows we highlight the parts that will be different from the inland case.

The weak formulation of the coastal generalized system will be different from the original inland version in that

- it will contain a boundary term for the wind because of the air - ocean interface,
- it will contain boundary terms for $\partial_2 C, \partial_3 C$, which had been assumed to be zero before on the ground.

The weak formulation is as follows.

Definition 3. *Find* $\mathbf{u}^\epsilon = (\mathbf{u}_H, u_3^\epsilon) \in L^2(0,T;\mathbf{V}) \cap L^\infty(0,T;L^2(\Omega)^3)$ *and* C^ϵ *with* $C^\epsilon \in L^\infty L^2$, $C^\epsilon \in L^2(0,T;H^1)$, $C^\epsilon(0) \in L^2_x$ *such that*

$$\begin{aligned}
&\int_0^T \Big\{ -(\mathbf{u}_H^\epsilon, \partial_t \tilde{\mathbf{u}}_H) + (\nabla_\nu \mathbf{u}_H^\epsilon, \nabla_\nu \tilde{\mathbf{u}}_H) - (\mathbf{u}_H^\epsilon, (\mathbf{u}^\epsilon \cdot \nabla)\tilde{\mathbf{u}}_H) + (b(\mathbf{u}_H^\epsilon), \tilde{\mathbf{u}}_H) \Big\}\, \mathrm{d}t \\
&+ \epsilon\beta \int_0^T \Big\{ (u_3^\epsilon, \tilde{u}_1) - (u_1^\epsilon, \tilde{u}_3) \Big\}\, \mathrm{d}t \\
&+ \epsilon^2 \int_0^T \Big\{ -(u_3^\epsilon, \partial_t \tilde{u}_3) + (\mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon, \tilde{u}_3) + (\nabla_\nu u_3^\epsilon, \nabla_\nu \tilde{u}_3) \Big\}\, \mathrm{d}t \\
&+ \int_0^T \Big\{ -(C^\epsilon, \partial_t \tilde{C}) - (\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}) + \epsilon K_1 (\partial_1 C^\epsilon, \partial_1 \tilde{C}) - \epsilon K_1 \langle \sigma_1, \tilde{C} \rangle_{\Gamma_G} + \\
&K_2(\partial_2 C^\epsilon, \partial_2 \tilde{C}) - K_2 \langle \sigma_2, \tilde{C} \rangle_{\Gamma_I} + K_3(\partial_3 C^\epsilon, \partial_3 \tilde{C}) - K_3 \langle \sigma_3, \tilde{C} \rangle_{\Gamma_I} \Big\}\, \mathrm{d}t \\
&= -(\mathbf{u}_{0H}^\epsilon, \tilde{\mathbf{u}}_H(0)) - \int_0^T \langle \theta_H, \tilde{\mathbf{u}}_H \rangle_{\Gamma_I} - \epsilon^2 (u_{03}^\epsilon, \tilde{u}_3(0)) \\
&- (C_0^\epsilon, \tilde{C}(0)) + \int_0^T (S^\epsilon, \tilde{C})\, \mathrm{d}t
\end{aligned} \tag{42}$$

for all $\tilde{\mathbf{u}} = (\tilde{\mathbf{u}}_H, \tilde{u}_3) \in H^1(0,T;\mathbf{V})$ *with* $\tilde{\mathbf{u}}(T) = 0$ *and* $\tilde{C}$ *such that* $\tilde{C} \in H^1(0,T,H^3_\Gamma(\Omega))$ *and* $\tilde{C}(T) = 0$.

On the other hand, the weak formulation of the hydrostatic case will take the following form.

Definition 4. *Find* $(\mathbf{u}, C)$ *such that* $\mathbf{u} = (\mathbf{u}_H, u_3) \in L^2(0,T;\mathbf{W})$ *with* $\mathbf{u}_H \in L^\infty(0,T;L^2(\Omega)^2)$, *and* $C \in L_t^\infty L_x^2$ *with* $\partial_2 C, \partial_3 C \in L_t^2 L_x^2$, *and* $C_0 \in L_x^2$, *such that*

$$
\begin{aligned}
&\int_0^T \Big\{ -(\mathbf{u}_H, \partial_t \tilde{\mathbf{u}}_H) + (\nabla_\nu \mathbf{u}_H, \nabla_\nu \tilde{\mathbf{u}}_H) - (\mathbf{u}_H, (\mathbf{u}\cdot\nabla)\tilde{\mathbf{u}}_H) + (b(\mathbf{u}_H), \tilde{\mathbf{u}}_H) \Big\}\,\mathrm{d}t \\
&+ \int_0^T \Big\{ -(C, \partial_t \tilde{C}) - (\mathbf{u}C, \nabla \tilde{C}) + \\
&K_2(\partial_2 C, \partial_2 \tilde{C}) - K_2\langle \sigma_2, \tilde{C}\rangle_{\Gamma_I} + K_3(\partial_3 C, \partial_3 \tilde{C}) - K_3\langle \sigma_3, \tilde{C}\rangle_{\Gamma_I} \Big\}\,\mathrm{d}t = \\
&= -(\mathbf{u}_{0H}, \tilde{\mathbf{u}}_H(0)) - \int_0^T \langle \theta_H, \tilde{\mathbf{u}}_H\rangle_{\Gamma_I} - (C(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C})\,\mathrm{d}t
\end{aligned} \tag{43}
$$

for all $(\tilde{\mathbf{u}}, \tilde{C})$ *with* $\tilde{\mathbf{u}} = (\tilde{\mathbf{u}}_H, \tilde{u}_3) \in H^1(0, T, \mathbf{W})$, *with* $\tilde{u}_H(T) = 0$ *and* $\partial_3 \tilde{\mathbf{u}}_H \in L^\infty(0, T; L^3(\Omega)^2)$ *and* $\tilde{C}$ *with* $\tilde{C} \in L^2(0, T, H^3_\Gamma(\Omega))$, $\tilde{C} \in H^1(0, T, L^2)$, $\tilde{C}(T) = 0$.

The coastal, generalized version of the energy inequality becomes

$$
\begin{aligned}
&\frac{1}{2}(\|\mathbf{u}^\epsilon_H(t)\|^2_{L^2} + \epsilon^2\|u^\epsilon_3(t)\|^2_{L^2} + \|C^\epsilon(t)\|^2_{L^2}) + \int_0^t \{\|\nabla_\nu \mathbf{u}^\epsilon_H\|^2_{L^2} + \epsilon^2 \|\nabla_\nu u^\epsilon_3\|^2_{L^2}\}\,\mathrm{d}\tau \\
&+ \int_0^t \{\epsilon K_1 \|\partial_1 C^\epsilon\|^2_{L^2} + K_2\|\partial_2 C^\epsilon\|^2_{L^2} + K_3\|\partial_3 C^\epsilon\|^2_{L^2}\}\,\mathrm{d}\tau \\
&\le \frac{1}{2}(\|\mathbf{u}^\epsilon_H(0)\|^2_{L^2} + \epsilon^2\|u^\epsilon_3(0)\|^2_{L^2} + \|C^\epsilon(0)\|^2_{L^2}) + \int_0^t (S^\epsilon, C^\epsilon)\,\mathrm{d}\tau \\
&- \int_0^t \langle \theta_H, \mathbf{u}_H\rangle_{\Gamma_I} + \int_0^t \epsilon K_1\langle \sigma_1, \sigma\rangle_{\Gamma_G} + K_2\langle \sigma_2, \sigma\rangle_{\Gamma_I} + K_3\langle \sigma_3, \sigma\rangle_{\Gamma_I}\,\mathrm{d}\tau.
\end{aligned} \tag{44}
$$

Since the σ boundary function is in L^2H^1, we have boundedness on the right hand side. The inequality provides the same estimates and weak convergence results. As for passing to the limit as $\epsilon \to 0$ in the coastal version of the anisotropic system seen above, there is no mathematical novelty compared to the inland case.

Eventually we arrive to the extended, coastal version of the main theorem, which is stated below.

Theorem 2. *Let* $\mathbf{u}_0 \in L^2(\Omega)^3$, *with* $\nabla\cdot\mathbf{u}_0 = 0, \mathbf{u}_0\cdot\mathbf{n} = 0$ *on* $\partial\Omega$; $C(0) \in L^2(\Omega)^3$, $C(0) = 0$ *on* $\partial\Omega$; *let* $\theta_1, \theta_2 \in L^2(0, T; H^{-1/2}(\Gamma_I))$ *and finally let* $\sigma \in L^2(0, T; H^1(\Gamma_G))$; *then as the aspect ratio* ϵ *tends to zero, any weak* *solution* $(\mathbf{u}^\epsilon, C^\epsilon)$ *of the coastal anisotropic equations (42) converge to a weak solution* $(\mathbf{u}$*,* $C)$ *of the hydrostatic coastal equations of the polluted atmosphere (43).*

Remark 4. *If we switch to the "informationless", fix coordinate system that does not match the downwind direction and we drop the assumption on the flow being convection dominated (ϵ - diffusion), then we can relax the restrictions on C^ϵ on the boundary. Indeed, in this case we do not have a vanishing ϵ term in front of the K_1 constant, and we can use the analogous estimate as in the inland case at the energy inequality and arrive to having a $(K_1 - \zeta\frac{K_1+K_2+K_3}{2}c_T(c_S+1))\|\partial_1 C^\epsilon\|^2_{L^2}$ term with a positive coefficient on the left hand side and a bounded $\frac{1}{\zeta_i}\frac{K_i}{2}\int_0^t \|\partial_i C^\epsilon\|^2_{L^2(\Gamma_G)}$ integral on the right hand side — the only assumption we need is $\partial_i C^\epsilon$ to be in $L^2_t L^2_x$ on the boundary.*

6 Concluding remarks

We are finishing this paper briefly mentioning a few ideas regarding the physical aspects of the model and some additional choices concerning the source term specifically.

Remark 5. In the paper [19] it is suggested that the domain we work on has the form of $p_0 < p < p_1$, because for the thin portion of air between the earth and the isobar $p = p_1$, another specific simplified model would be necessary. This means that a way to make the model more precise is to cut the domain at an isobar a little bit off the physical ground. This would of course radically change the boundary conditions we use for the functions (in fact it is not trivial how to change them for this case), because it would take away the advantage of having a wall-like, natural and physical boundary condition for the domain we work on. If we are positively above the ground, $\partial_3 C = 0$ would represent a strange reflective layer.

Remark 6. Note that the aspect ratio going to zero is not only legitimate on a global scale. When we switch from a global model to a local one in order to be able to use the beta plane approximation, we still have hundreds of thousands of square kilometers versus 1—5 kilometers.

Remark 7. We chose the Gaussian nascent delta function to give shape to the source term, but it can be defined in several different ways using the many different pulses that approximate the delta function (choosing only those however that are physically meaningful in our case, for example its values are nonnegative), e.g.

a) the unit impulse: $\delta_\epsilon(\mathbf{x}-\mathbf{x_s}) = \epsilon/2, \big|(\mathbf{x}-\mathbf{x_s})\big| < 1/\epsilon$,

b) the Lorentzian pulse: $\delta_\epsilon(\mathbf{x}-\mathbf{x_s}) = \epsilon/(1+\pi^2\epsilon^2(\mathbf{x}-\mathbf{x_s})^2)$.

The Gaussian and Lorentzian version of the source term is continuous and bounded on Ω, it is actually in L^∞ in space (and in L^2 as a consequence), while it is easy to see that the piecewise constant unit pulse is in L^2 as well. Since the time dependence of this source is described by the Heaviside function, the time norm is also finite.

A The existence of weak solution of the hydrostatic system of the polluted atmosphere

In this section for completeness we sketch the proof of the existence of weak solutions of the system (18) - (25).

The idea is that the polluted atmosphere can be viewed as the air-analogous version of the salty sea water in terms of the equations that represent it. In both cases we have a fluid which can be seen approximately as a homogenous incompressible fluid carrying some other particles in a significant concentration, the basis fluid motion is well described by the incompressible Navier-Stokes equations, and the additional concentration is described by an advection-diffusion equation.

The paper of [19] provides an existence result for an arbitrary time interval $[0, t_1]$ for the system describing the salty ocean water with temperature. The structure of the equations themselves describing the system is essentially the same, if we drop the equation describing the temperature, we basically arrive to the same system that describes the polluted atmosphere.

The main difference consists in the boundary conditions: on every boundary they use a zero flux for the salinity, $\frac{\partial S}{\partial \mathbf{n}_s} = 0$. This coincides with our lower boundary conditions, but on the lateral boundary we use a fix constant.

We will follow step by step the structure of the proof in [19], redefining the necessary quantities to be valid for our case and verify that the proof holds in the case of the polluted atmosphere as well (this is necessary not only because of using altered boundary conditions but also because the regularities of the functions are different in some cases).

Step 1. Firstly we redefine the norms and the functional setting in order to make them match our system.

Throughout this section we will use the $V = V_1 \times V_2, H = H_1 \times H_2, \mathcal{V} = \mathcal{V}_1 \times \mathcal{V}_2$, and $V_{(2)}$ function spaces.

We denote by U the vector $(\mathbf{u}_H, C)$. The scalar products and norms in the case of the polluted atmosphere are defined as follows.

$$((U,\tilde{U})) = ((\mathbf{u}_H, \tilde{\mathbf{u}}_H))_1 + K_C((C,\tilde{C}))_2$$

$$((\mathbf{u}_H, \tilde{\mathbf{u}}_H))_1 = \int_\Omega \nabla_\nu \mathbf{u}_H \nabla_\nu \tilde{\mathbf{u}}_H \,\mathrm{d}\omega$$

$$((C,\tilde{C}))_2 = \int_\Omega K_2 \partial_2 C \partial_2 \tilde{C} + K_3 \partial_3 C \partial_3 \tilde{C} \,\mathrm{d}\omega$$

$$\|U\| = ((U,U))^{1/2}$$

$$(U,\tilde{U})_H = \int_\Omega -\mathbf{u}_H \tilde{\mathbf{u}}_H - K_C C\tilde{C} \,\mathrm{d}\omega$$

$$|U|_H = \left|(U,U)_H\right|^{1/2}$$

Step 2. Now the hydrostatic weak formulation of the polluted atmosphere can be written as

$$\begin{aligned} &\int_0^T (U, \frac{\mathrm{d}}{\mathrm{d}t}\tilde{U})_H \,\mathrm{d}t + \int_0^T a(U,\tilde{U}) + b(U,U,\tilde{U}) + e(U,\tilde{U}) \,\mathrm{d}t = \\ &= \int_0^T l(\tilde{U}) \,\mathrm{d}t + \int_0^T \int_\Omega U_0 \tilde{U}(0) d\omega \,\mathrm{d}t \end{aligned} \tag{45}$$

with

$$a(U,\tilde{U}) = a_1(U,\tilde{U}) + K_C a_2(U,\tilde{U})$$

$$a_1(U,\tilde{U}) = \int_\Omega \nabla_\nu \mathbf{u}_H \nabla_\nu \tilde{\mathbf{u}}_H \,\mathrm{d}\omega$$

$$a_2(U,\tilde{U}) = \int_\Omega K_2 \partial_2 C \partial_2 \tilde{C} + K_3 \partial_3 C \partial_3 \tilde{C} \,\mathrm{d}\omega$$

$$b = b_1 + K_2 b_2$$

$$b_1(U,\tilde{U},U^{\#}) = -\int_\Omega (\mathbf{u}_H \tilde{\mathbf{u}}_H + \mathbf{u}_H \tilde{u}_3) \nabla \mathbf{u}_H^{\#}$$

$$b_2(U,\tilde{U},U^{\#}) = -\int_\Omega (\mathbf{u}_H\tilde{C} + u_3\tilde{C})\nabla C^{\#}$$

$$e(U,\tilde{U}) = \int_\Omega b(\mathbf{u}_H)\tilde{\mathbf{u}}_H \,\mathrm{d}\omega$$

$$l(\tilde{U}) = -\int_\Omega S\tilde{C}\,\mathrm{d}\omega.$$

Here the term $b(\mathbf{u}_H) = \alpha(-u_2, u_1)$ represents the cross product, and we use the fact that the vertical velocity w can be expressed as $u_3 = u_3(\mathbf{u}_H)$ because the divergence of the three dimensional velocity is zero.

Step 3. It is straightforward to see that a is trilinear continuous, the coercivity of a is also trivial, moreover e is bilinear continuous and $e(U,U) = 0$.

What changes slightly is the proof of the results available on the expression b since the space derivatives in our case are applied on the test function.

- To prove the bound for $U,\tilde{U} \in V, U^{\#} \in V_{(2)}$, in this case we can not use the L_∞ bound on the third term, since we would eventually arrive to $\|C^{\#}\|_{H^3}$, which means that we can not close the bound that way. We can use

$$\int_\Omega u_3\tilde{C}\nabla C^{\#} d\omega \le c\|u_3\|_{L^2}\left\|\tilde{C}\right\|_{L^4}\left\|\nabla C^{\#}\right\|_{L^4} \le$$
$$\le c\|U\|\left\|\tilde{U}\right\|\left\|\nabla C^{\#}\right\|_{H^1} \le c\|U\|\left\|\tilde{U}\right\|\left\|U^{\#}\right\|_{V^{(2)}}.$$

- For $U, U^{\#} \in V, \tilde{U} \in V_{(2)}$ we have

$$\int_\Omega u_3\tilde{C}\nabla C^{\#} d\omega \le c\left\|\tilde{C}\right\|_{L^\infty}\|u_3\|_{L^2}\left\|\nabla C^{\#}\right\|_{L^2} \le c\|U\|\left\|\tilde{U}\right\|_{V^{(2)}}\left\|U^{\#}\right\|.$$

- The proof of $b(U,\tilde{U},U^{\#}) = -b(U,U^{\#},\tilde{U})$ can be derived similarly as we have

$$\int_0^T (\mathbf{u}\nabla C, \tilde{C})\,\mathrm{d}t = -\int_0^T (-\mathbf{u}C, \nabla\tilde{C})\,\mathrm{d}t,$$

observing that the boundary terms disappear.

- The establishment of the improvement

$$|b(U,\tilde{U},U^{\#})| \leq c\|U\|\,|\tilde{U}|_H^{1/2}\|\tilde{U}\|^{1/2}\|U^{\#}\|_{V_{(2)}}$$

of the first bound for $U, \tilde{U} \in V, U^{\#} \in V_{(2)}$, we consider the most typical term and we bound it by

$$\int_\Omega u_3 \tilde{C} \nabla C^{\#} \,\mathrm{d}\omega \leq \|u_3\|_{L^2} \left\|\tilde{C}\right\|_{L^3} \left\|\nabla C^{\#}\right\|_{L^6}.$$

Note that since our definition of the form b is different from [19], thus we need to skip the switch of function arguments caused by the application of $|b(U,\tilde{U},U^{\#})| = |b(U,U^{\#},\tilde{U})|$, otherwise the bound can not be closed.

Now, using the $\|\varphi\|_{L^3} \leq c\|\varphi\|_{L^2}^{1/2}\|\varphi\|_{H^1}^{1/2}$ Sobolev-Ladyzhenskaya inequality in space dimension 3, we can bound this term by

$$c\|\mathbf{u}_H\| \left\|U^{\#}\right\|_{V_{(2)}} |\tilde{U}|^{1/2}\left\|\tilde{U}\right\|^{1/2},$$

which yields the inequality.

This means that all the necessary properties of the forms a, b and e are equally valid for our set of equations (18) - (25), and we can use them to prove a stationary-case existence theorem identically as the proof is constructed in [19].

The weak formulation is as follows.

Definition 5. *Given a source term $S = \delta$ or $S \in L^2(\Omega)$, find $U = (\mathbf{v}, C)$, that satisfies the hypothesis of Definition 1 in the space dimensions, moreover*

$$a(U,\tilde{U}) + b(U,U,\tilde{U}) + e(U,\tilde{U}) = l(U,\tilde{U}) \tag{46}$$

for every $\tilde{U} \in V_{(2)}$.

Note that the definition of the weak formulation in our case is a bit more general, since we allow not only L^2 functions to represent the source term, but the δ distribution as well. As we have seen before, the weak formulation is still valid in this case because of the basic properties of δ, so this generalization is mathematically valid.

We have the following theorem establishing the existence of solutions of the stationary equations, which can be proved by Galerkin method using the fundamental results we verified for the functions a, b and e.

Theorem 3. *We are given $S = \delta$ or S in $L^2(\Omega)$, then the problem (46) possesses at least one solution $U \in V$ such that*

$$\|U\| \leq \frac{1}{c_1}\|l\|_{V'},$$

where c_1 is the coercivity constant.

Step 4. In this step we provide the operator form of the equation (45) in the Hilbert space $V'_{(2)}$.

The operators A, B and E can be set analogously as in the original version except for the additional integration in the time dimension; A for example is defined to be a linear continuous form from V into V', given by the formula

$$< AU, \tilde{U} >= \int_0^T a(U, \tilde{U})\,\mathrm{d}t, \quad \forall U, \tilde{U} \in V.$$

The main difference in this part of the proof is that we will need an additional operator Z for the time derivative part, since we have to capture the

$$\int_0^T (U, \frac{\mathrm{d}}{\mathrm{d}t}\tilde{U})_H\,\mathrm{d}t$$

inner product in terms of operators. In the original version $\frac{\mathrm{d}U}{\mathrm{d}t}$ was used as an operator acting on $\tilde{U}$, the result defined to be their inner product. In our case, since we apply the weak formulation in space-time, and the time derivation is applied on the test function, we introduce the following operator Z:

$$< Z(U), \tilde{U} >= \int_0^T (U, \frac{d\tilde{U}}{dt})_H\,\mathrm{d}t.$$

With this, the operator form of the equation becomes

$$ZU + AU + B(U,U) + EU = l + U_0.$$

Step 5. Finally we discuss the details of the time-dependent case, where we establish the existence, for all time, of the solutions for the system (18)-(22).

The weak formulation in the time-dependent case takes the following form.

Definition 6. *Given T, $U_0 \in H$, and a source term $S = \delta$ or $S \in L^2(\Omega)$, find U that satisfies the conditions of (26), moreover*

$$\begin{aligned}\int_0^T (U, \frac{\mathrm{d}}{\mathrm{d}t}\tilde{U}) + a(U,\tilde{U}) + b(U,U,\tilde{U}) + e(U,\tilde{U})\,\mathrm{d}t = \\ \int_0^T l(\tilde{U})\,\mathrm{d}t + (U_0, \tilde{U}(0)), \quad \forall \tilde{U} \in V_{(2)}\end{aligned} \tag{47}$$

We proceed by using finite differences in time, taking an arbitrary integer N, setting a time step $k = \Delta t = T/N$, and recursively for $n = 1, ..., N$ considering

$$\begin{aligned} U^0 &= U_0, \\ -\frac{1}{\Delta t}(U^n - U^{n-1}, \tilde{U})_H &+ a(U^n,\tilde{U}) + b(U^n,U^n,\tilde{U}) \\ &+ e(U^n,\tilde{U}) = l^n(\tilde{U}), \quad \text{for any } \tilde{U} \in V_{(2)}\end{aligned} \tag{48}$$

We use $-\frac{1}{\Delta t}(U^n - U^{n-1}, \tilde{U})_H$ to approximate $(U, \frac{\mathrm{d}}{\mathrm{d}t}\tilde{U})$, and note that we will have a minus sign difference from the original version — as seen in the above expression — since we apply the time derivative on the test function.

Following the proof of the existence of $U^n \in V$, proving some a priori estimates, introducing approximate functions on (0,T), and using the Aubin compactness theorem in an identical way as described in [19], we pass to the limit and arrive to

$$\int_0^T (U,\tilde{U})_H \psi' \,\mathrm{d}t + \int_0^T \left[a(U,\tilde{U}) + b(U,U,\tilde{U}) + e(U,\tilde{U})\right]\psi \,\mathrm{d}t =$$

$$= (U_0,\tilde{U})\psi(0) + \int_0^T l(\tilde{U})\psi\,\mathrm{d}t,$$

where ψ is a function defined on $[0, T]$, used to create the time-dependent version $\tilde{U}\psi$ of the test function. Renaming this product simply to be $\tilde{U}(t)$, the time-dependent test function, we arrive exactly to

$$\int_0^T (U, \partial_t\tilde{U})_H \,\mathrm{d}t + \int_0^T \left[a(U,\tilde{U}) + b(U,U,\tilde{U}) + e(U,\tilde{U})\right]\mathrm{d}t =$$

$$= (U_0, \tilde{U}(0)) + \int_0^T l(\tilde{U})\,\mathrm{d}t,$$

which is the same as (47), and which thus finally leads us to the main existence result of this section.

Theorem 4. *Given $T > 0$, $U_0 \in H$, and a source term $S = \delta$ or $S \in L^2(\Omega)$, then there exists $(\mathbf{u}, C)$ with $u_3 = -\int_0^T \nabla \cdot \mathbf{u}_H$, $\mathbf{u} = (\mathbf{u}_H, u_3) \in L^2(0, T; \mathbf{W})$ with $\mathbf{u}_H \in L^\infty(0, T; L^2(\Omega)^2)$, and $C \in L_t^\infty L_x^2$ with $\partial_2 C, \partial_3 C \in L_t^2 L_x^2$, which is solution to (26).*

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